KENNESAW STATE
UNIVERSITY

CS 4277
Logistic/Softmax Regression

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Computer Science

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Learning Objectives

- Describe logistic regression algorithm
- Solve classification problems using logistic regression
- Describe the loss function of logistic regression
- Solve multi-class classification problems using logistic regression

Financial fraud

- Name
- Description
- Account
- Sender
- Receiver
- …

Customer analysis

- User
- Gender
- Age
- Salary
- …

Disease detection (e.g. COVID-19)

- User
- Gender
- Age
- Lifestyle
- Risk

…

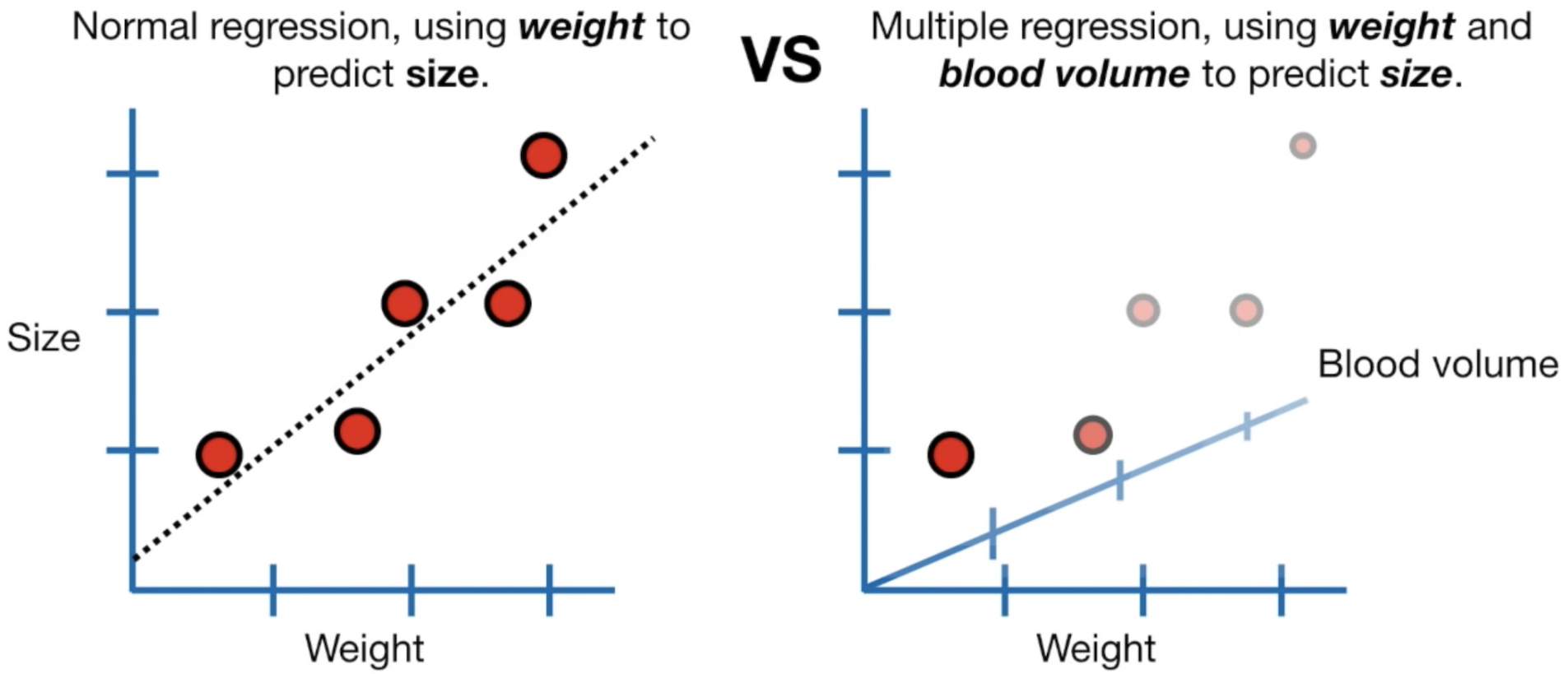
Logistic Regression

Logistic Regression

- Predict results on a binary outcome variable
- E.g.,
  - Whether or not a patient has a disease
  - Whether a facial image is male or female

- The outcome is not continuous or distributed normally

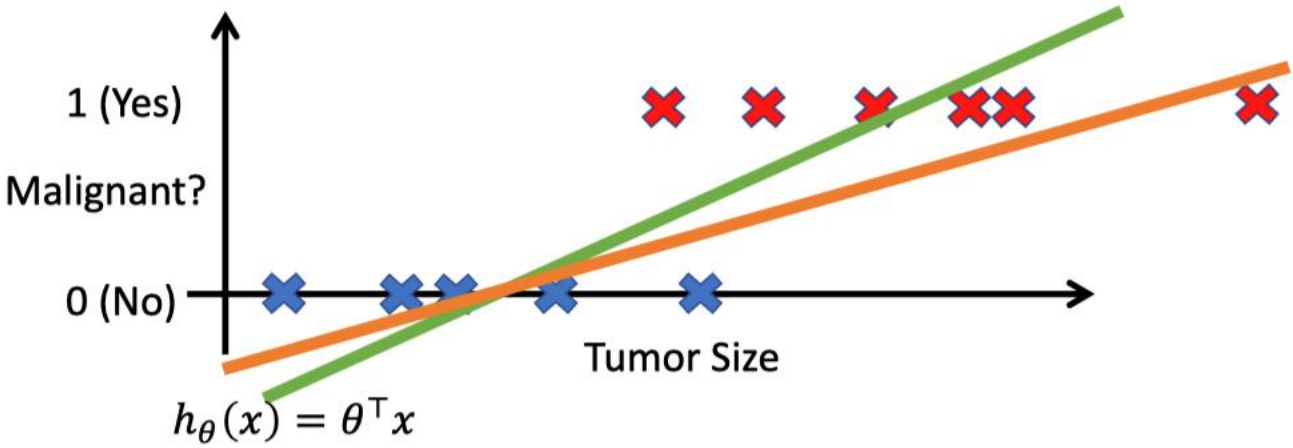
Recall: Linear Regression



- Comparing the simple model to the complicated one tells us if we need to measure weight and blood volume to accurately predict size or if we get away with just weight

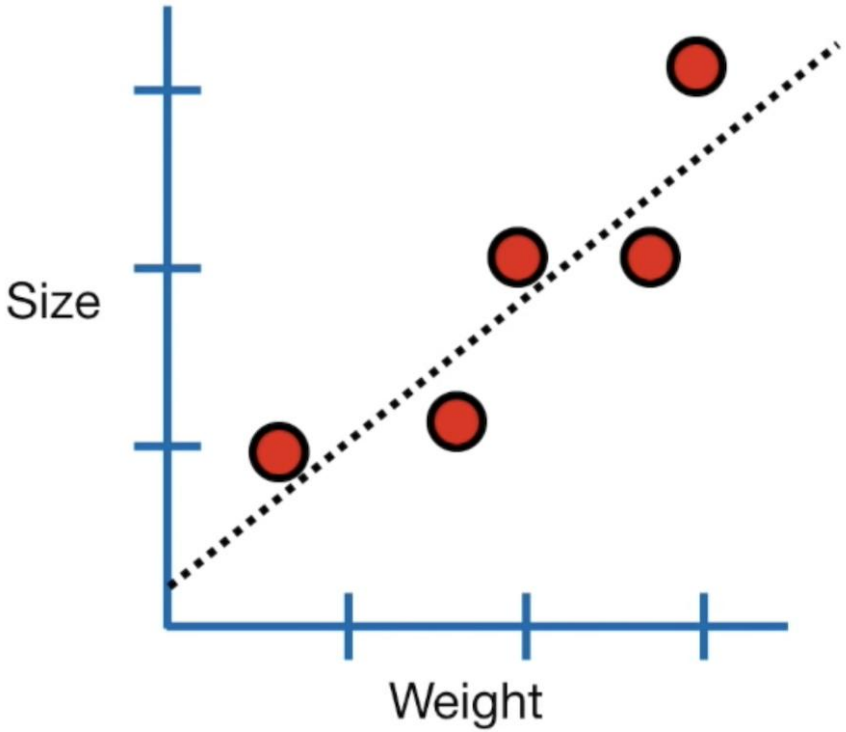
Binary Classification

- Can we use linear regression?
- Threshold classifier output $h_\theta(x)$ at 0.5
- If $h_\theta(x) \geq 0.5$, predict " $y = 1$"
- If $h_\theta(x) < 0.5$, predict " $y = 0$"
- Problem 1: if we had a single Yes with a very large tumor.
- Problem 2: Hypothesis can give values large than 1 or less than 0.

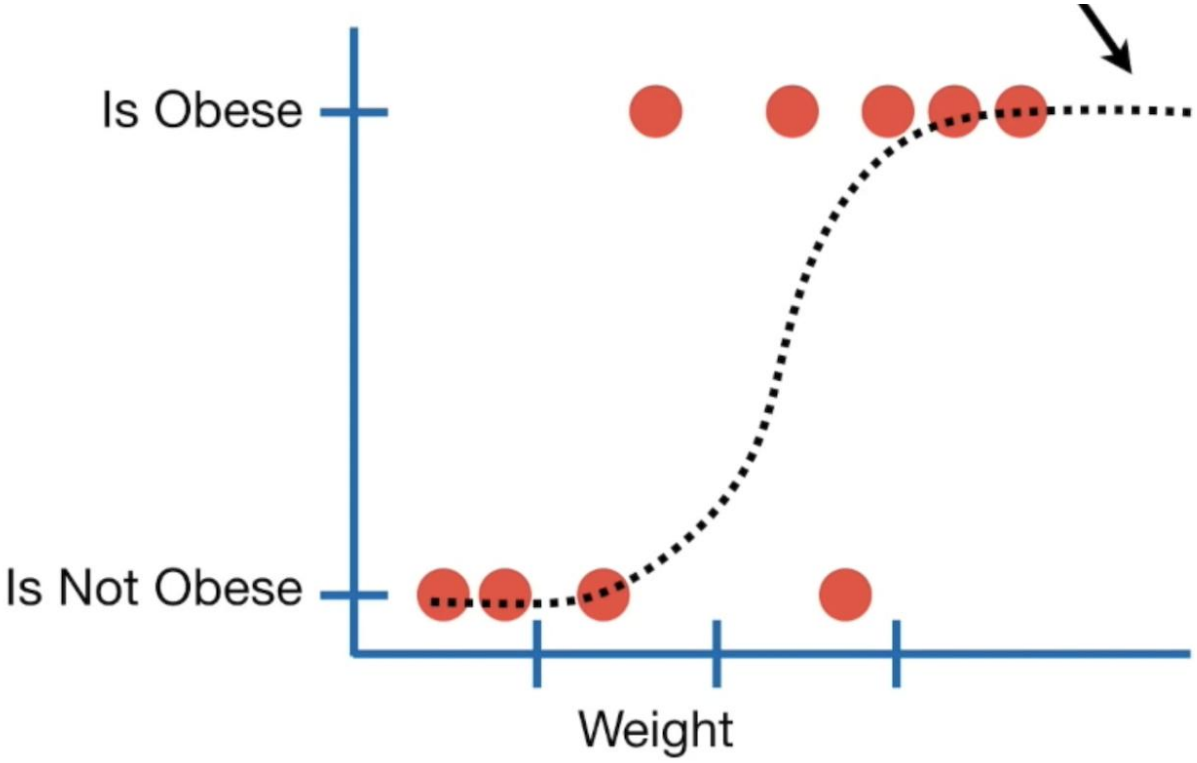


Logistic regression

- Logistic regression is similar to linear regression, excepts it predicts something is true or false

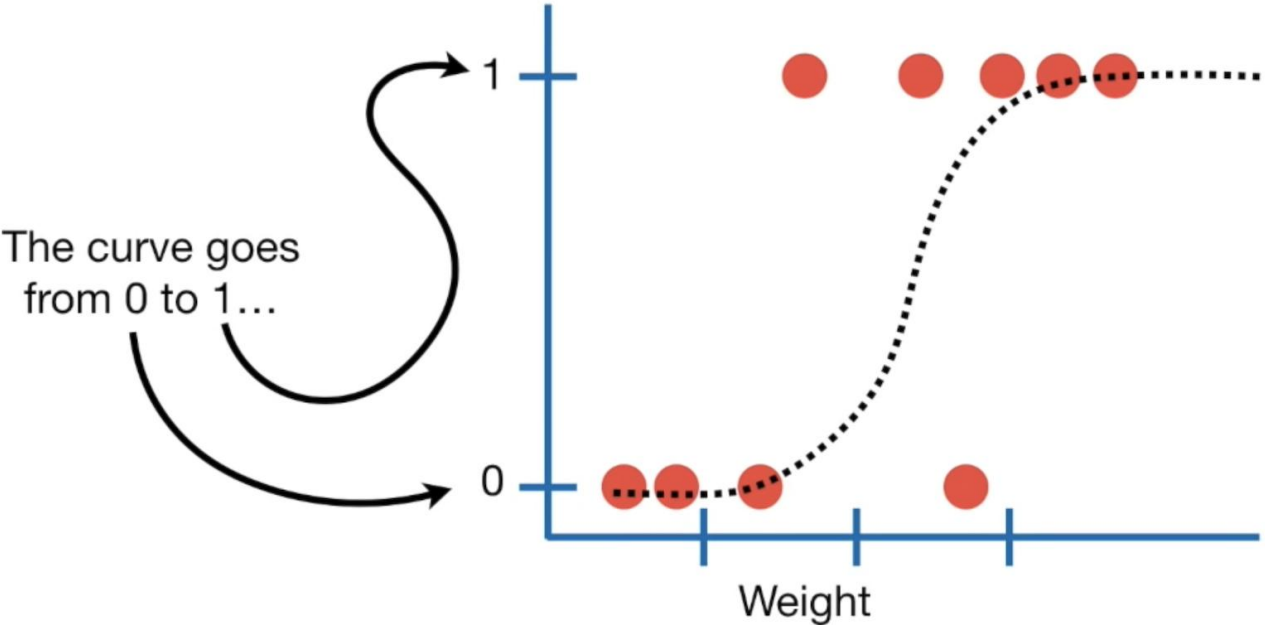


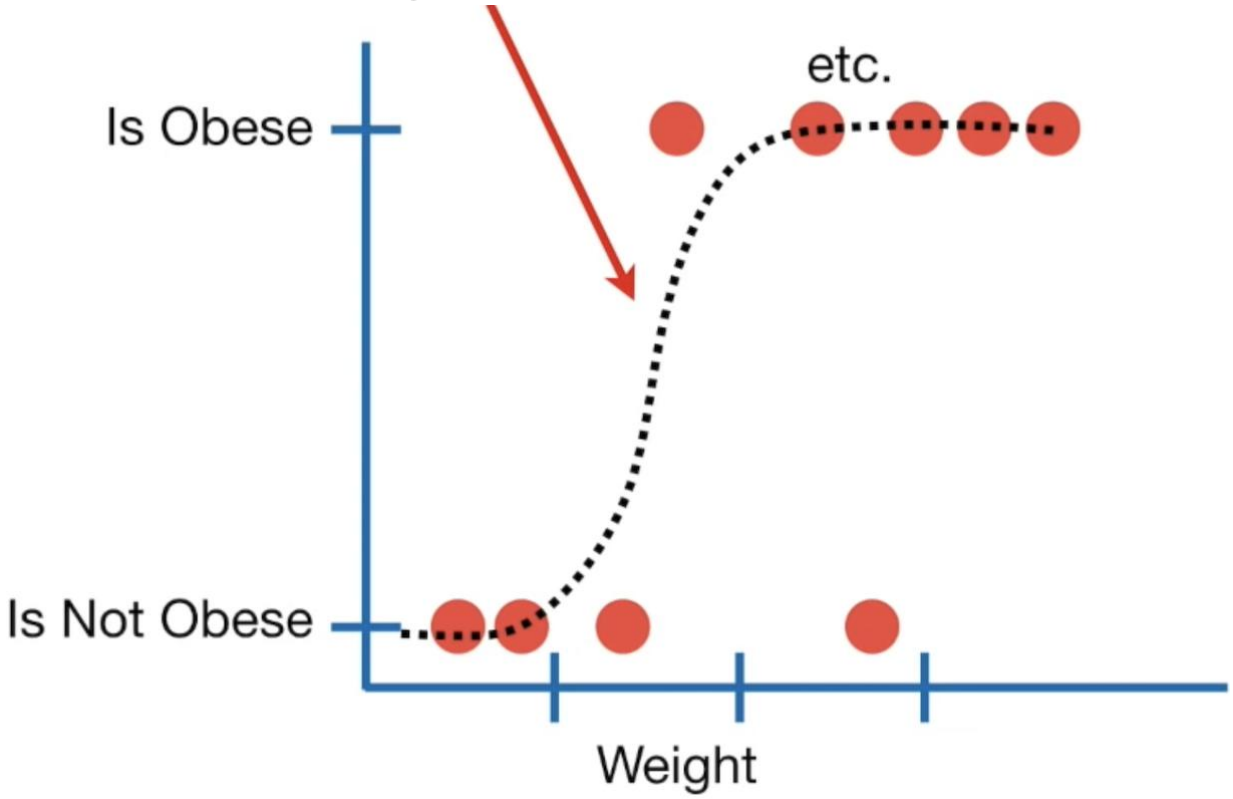
- Instead of fitting a line to a the data, logistic regress fits an “S” shaped “logistic function”



Logistic regression

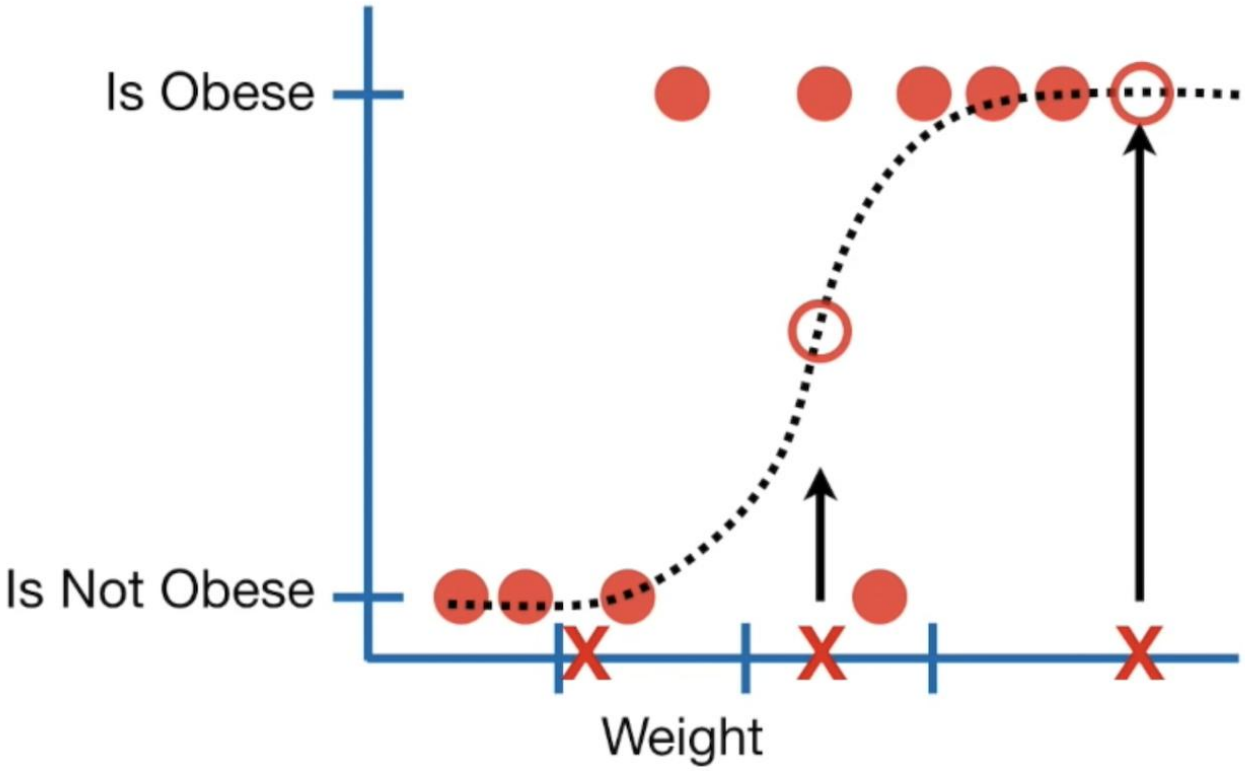
- The curse tells you the probability that a mouse is obese based on its weight



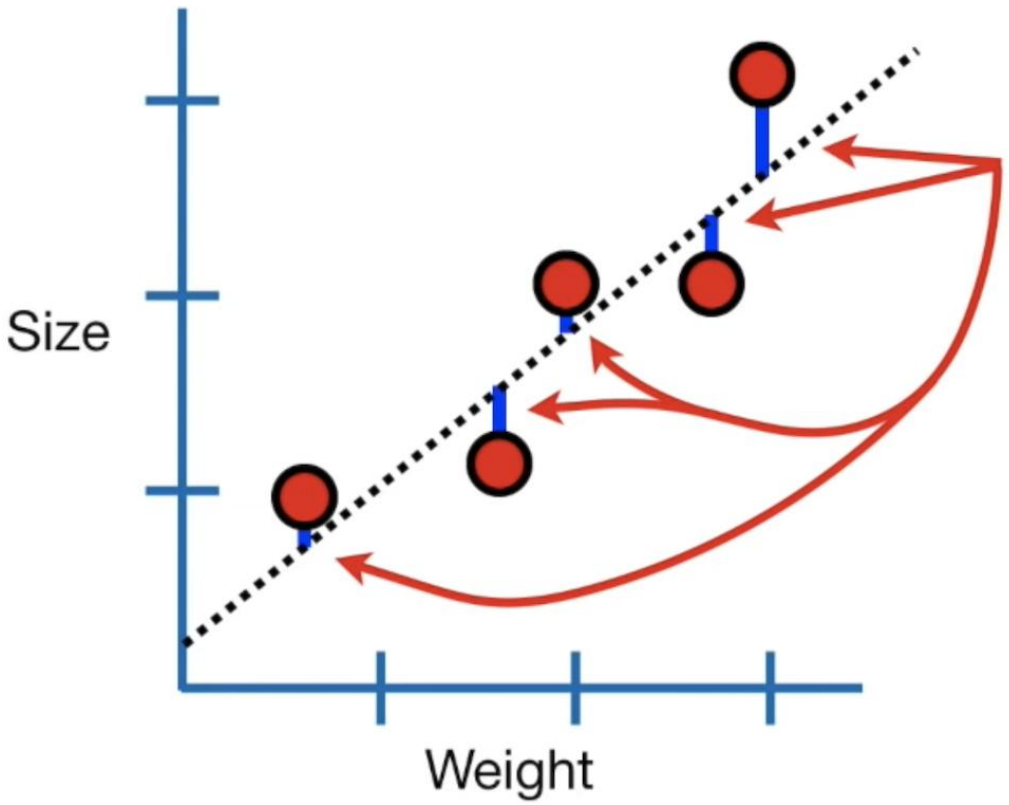


Logistic regression

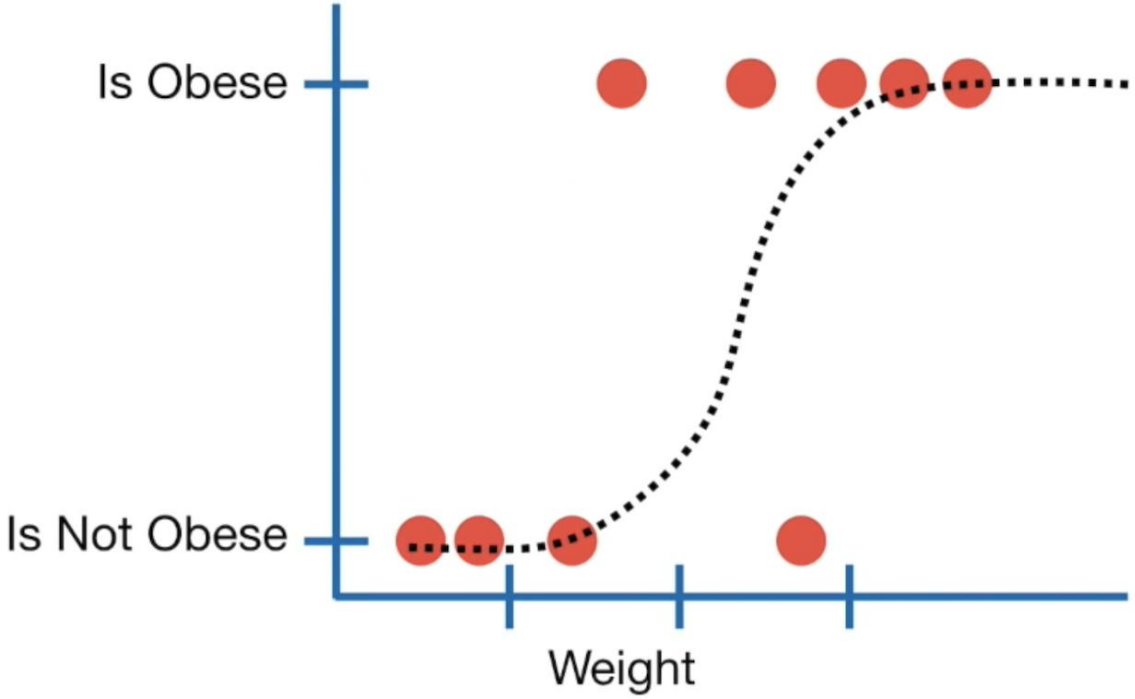
- Logistic regression's ability to provide probabilities and classify new samples using Continuous and discrete measurements makes it A popular machine learning method



Logistic regression vs Linear Regression



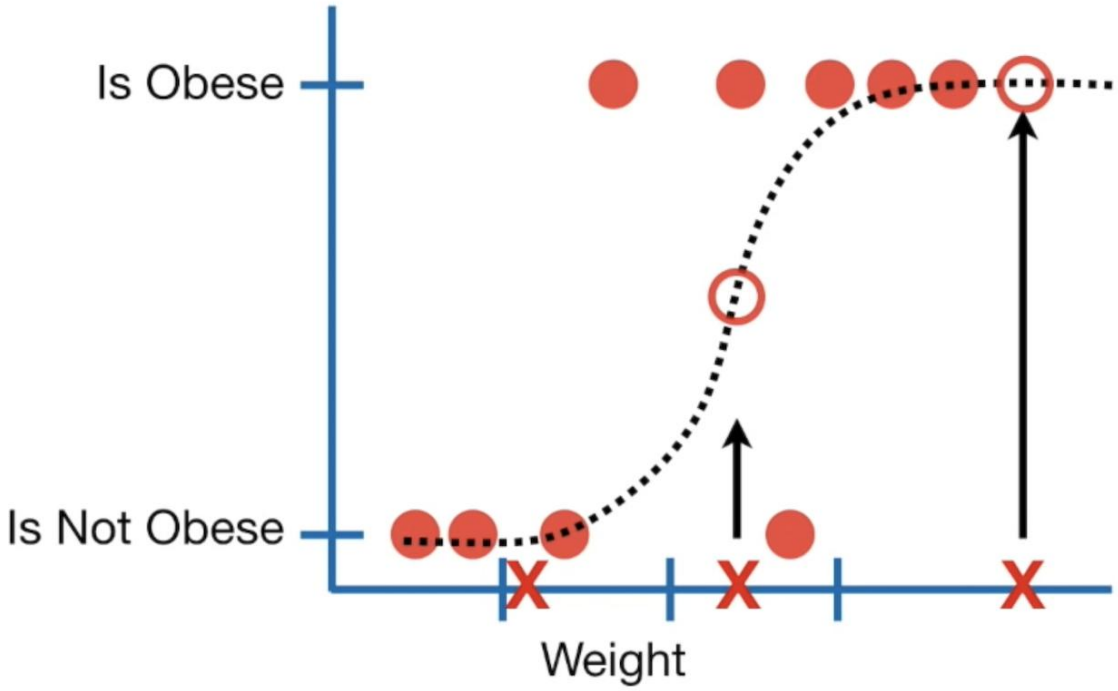
- We find a line that minimizes the sum of square of these residuals



- Logistic regression does not have same concept as residuals, so it can't use least square.
- Instead, it uses maximum log likelihood

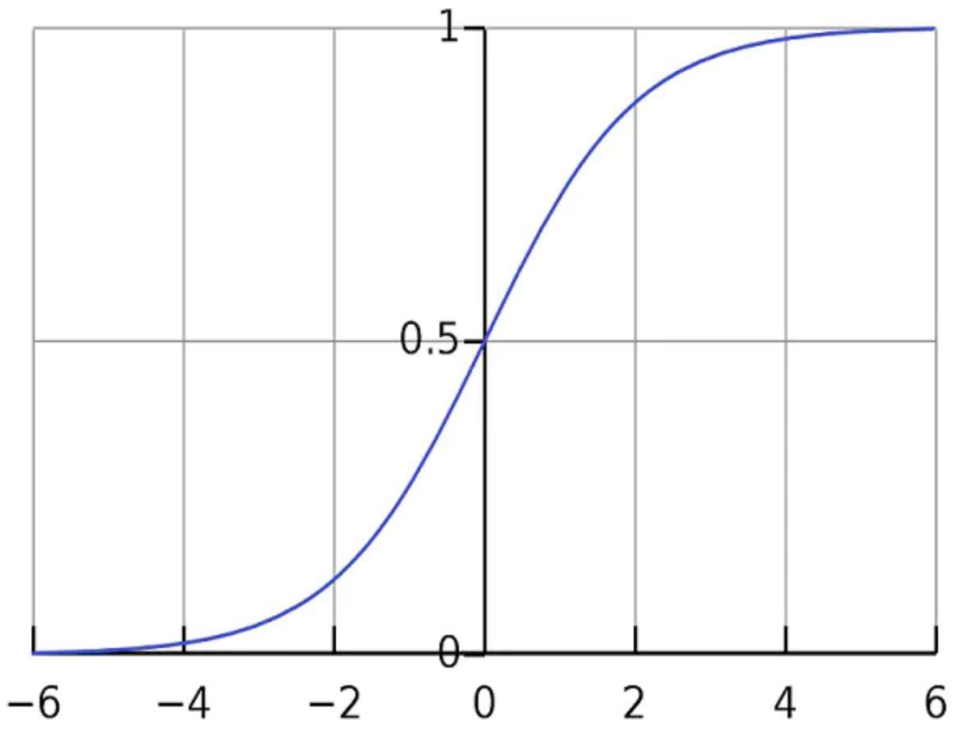
Logistic regression

- Calculate likelihood for all mice/data points.
- Then multiply all these likelihood together to get the likelihood of the data.
- ”S” shaped (squiggle” gives the likelihood of all the data together.



Sigmoid function

- Definition: $g(x) = \frac{1}{1+e^{-x}}$
- Sigmoid Function can map linear values to the range [0-1].
- If the mapping result is less than 0.5, it is considered a negative sample.
- If it is greater than 0.5, it is considered a positive sample.



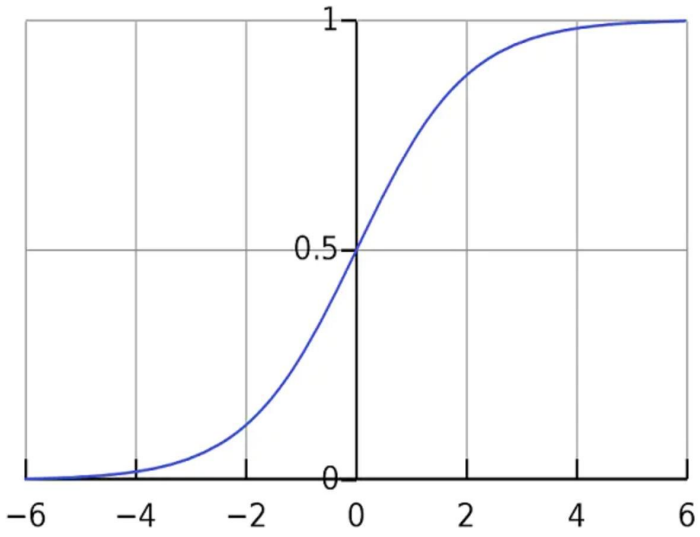
Logistic regression vs. Linear regression

- Linear regression:

$$z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots = \theta^T x$$

- Logistic regression:

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{1}{1 + e^{-z}}$$

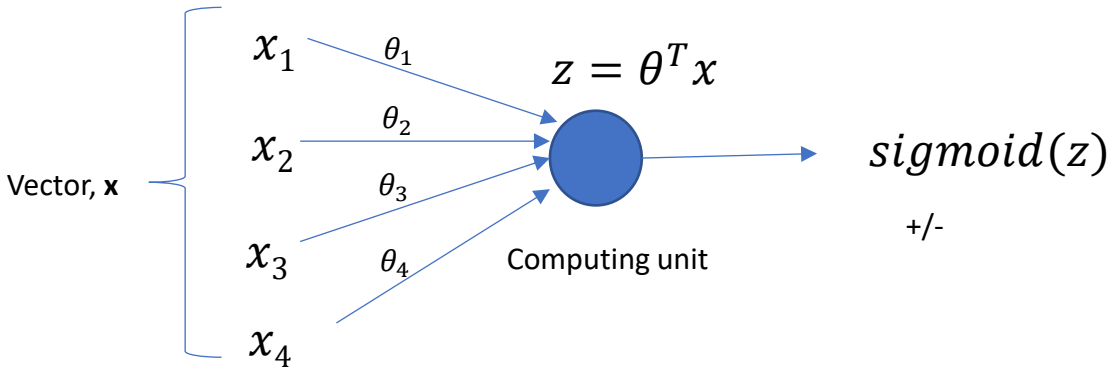


$z = \theta^T x$

- Suppose predict "$y = 1$" if $h_\theta(x) \geq 0.5$, $z = \theta^T x \geq 0$
- Suppose predict "$y = 0$" if $h_\theta(x) < 0.5$, $z = \theta^T x < 0$

Computing Unit

- Single computational unit
  - Input vector/feature,
  - Weights,
  - Computing unit/functions
  - Output

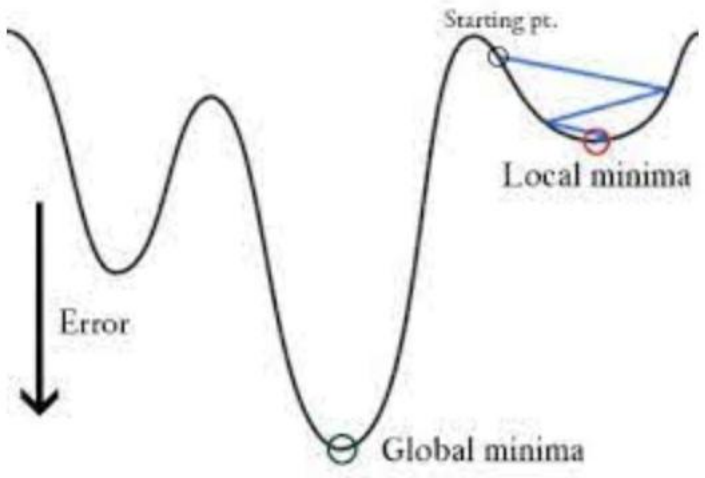


- Takes real valued numbers as input → Perform Computation → output

Cost function & Optimization

Can we use ”mean square error” as cost for Logistic regression

- Logistic Regression is a nonlinear function ($h_\theta(x) = \frac{1}{1+e^{-z}}$)
  - if we put this in the above MSE equation it will give a non-convex function as shown:



Can we use ‘mean square error’ as cost for Logistic regression

- When we try to optimize values using gradient descent it will create complications to find global minima.
- Another reason is in classification problems, we have target values like 0/1, So $(\hat{Y}-Y)^2$ will always be in between 0-1 which can make it very difficult to keep track of the errors and it is difficult to store high precision floating numbers.
- Sensitivity to Outliers: MSE is sensitive to outliers, which may not be suitable for classification problems where the focus is on predicting class membership probabilities.
- *The cost function used in Logistic Regression is **Log Loss**.*

What is Log Loss?

- Log Loss is the most important classification metric based on probabilities
- It's hard to interpret raw log-loss values, but log-loss is still a good metric for comparing models.
- For any given problem, a lower log loss value means better predictions.
- *Mathematical interpretation:*
  - Log Loss is the negative average of the log of corrected predicted probabilities for each instance.

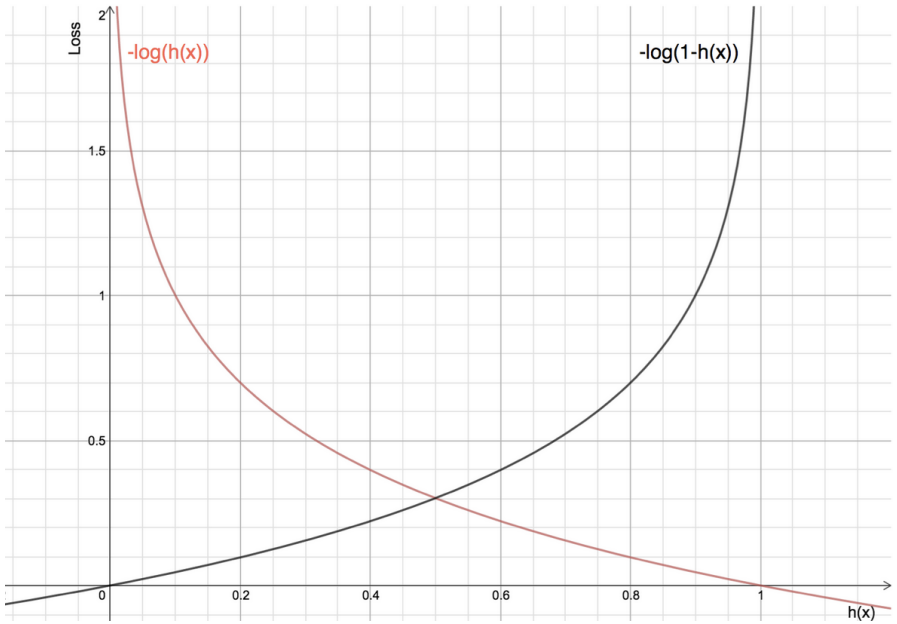
What is Log Loss?

- The model is giving predicted probabilities

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{1}{1 + e^{-z}}$$

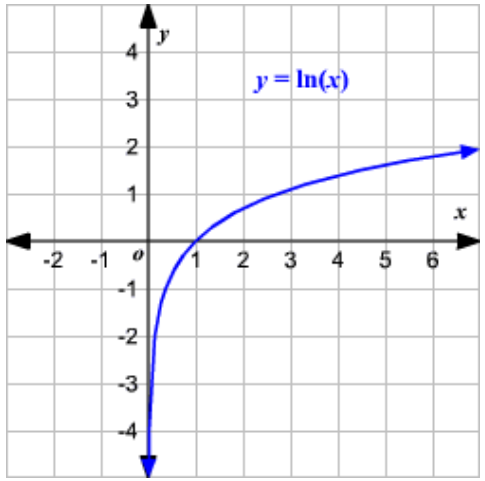
- log loss measures the UNCERTAINTY of the probabilities of LR model by comparing them to the true labels.
  - your training labels are 0 and 1 but your training predictions are 0.4, 0.6, 0.89, 0.1122 etc.
- How do we calculate a measure of the error of our model ?
  - If we directly classify all the observations having values > 0.5 into 1 then we are at a high risk of increasing the misclassification.
  - Many values having probabilities 0.4, 0.45, 0.49 can have a true value of 1.

The cost function graphs for actual class 1 and 0



- If $y = 1$, cost=$-\log(h_\theta(x))$
- If $y = 0$, cost=$-\log(1 - h_\theta(x))$
- $Cost(h_\theta(x), y) = -y\log\big(h_\theta(x)\big) - (1 - y)\log(1 - h_\theta(x))$
- Binary Cross-Entropy / Log Loss

Log-loss



- Address the issues: Log loss came to the scenario

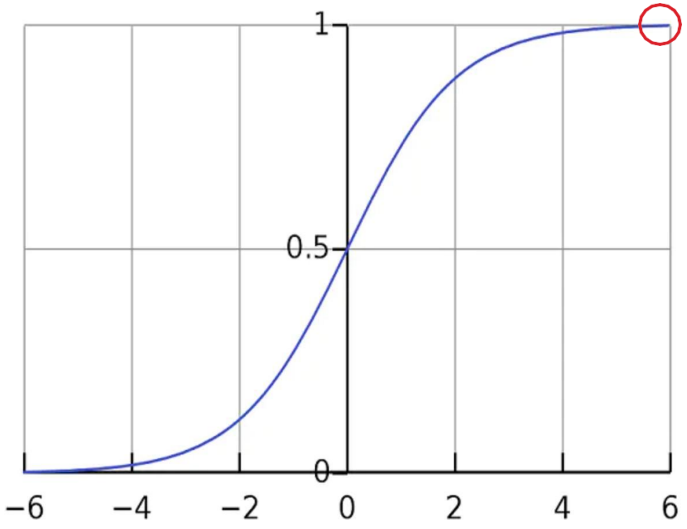
$$logloss = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(h_\theta(x_i)\right) + (1 - y_i)\log\left(1 - h_\theta(x_i)\right)\right]$$

- Case 1 : y_i=1 ,$h_\theta(x_i)$ = High, 1−y_i=0 , 1−$h_\theta(x_i)$ = Low
- Case 2 : y_i=1 ,$h_\theta(x_i)$ = Low , 1−y_i=0 , 1−$h_\theta(x_i)$ = High
- Case 3 : y_i=0 ,$h_\theta(x_i)$ = Low , 1−y_i=1 , 1−$h_\theta(x_i)$ = High
- Case 4 : y_i=0 ,$h_\theta(x_i)$ = High , 1−y_i=1 , 1−$h_\theta(x_i)$ = Low

Cost function of logistic regression

- $logloss = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(h_\theta(x_i)\right) + (1-y_i)\log\left(1-h_\theta(x_i)\right)\right]$
- Case 1 : y_i=1, $h_\theta(x_i)$ = High, 1−y_i=0 , 1−$h_\theta(x_i)$ = Low

It means no cost if predicted value is the same as the original value



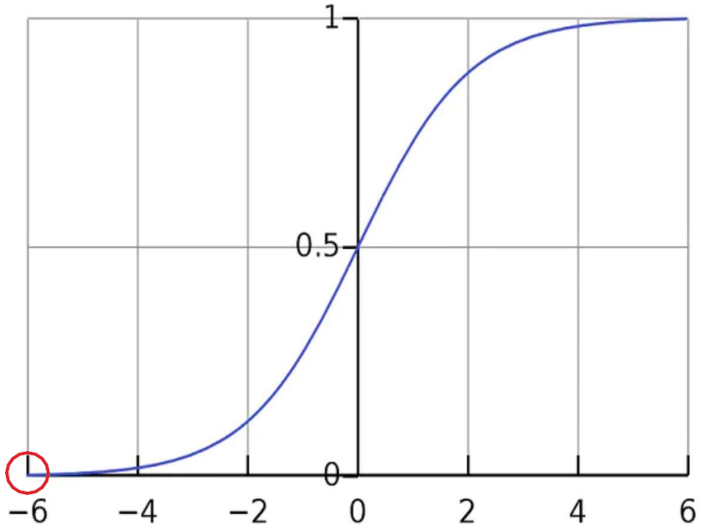
y_i=1

Cost function of logistic regression

- $logloss = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(h_\theta(x_i)\right) + (1-y_i)\log\left(1-h_\theta(x_i)\right)\right]$
- Case 3 : y_i=0 ,$h_\theta(x_i)$ = Low , 1−y_i=1 , 1−$h_\theta(x_i)$ = High

It means no cost if predicted value is the same as the original value

y_i=0



Cost function of logistic regression

- $logloss = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(h_\theta(x_i)\right) + (1 - y_i)\log\left(1 - h_\theta(x_i)\right)\right]$
- The more consistent the original value is with the predicted result, the smaller the loss function will be; otherwise, the loss function will be larger.

Cost function of logistic regression

- Suppose the probability of the classification result for x is 1:

$$P(y=1|x;\theta) = h_\theta(x)$$
$$P(y=0|x;\theta) = 1 - h_\theta(x)$$

- Combine these two terms together as:

$$P(y|x;\theta) = \left(h_\theta(x)\right)^y \left(1 - h_\theta(x)\right)^{1-y}$$

- We can get the likelihood function for a set of training samples

$$L(\theta) = \prod_{i=1}^{n} P(y_i|x_i;\theta) = \prod_{i=1}^{n} \left(h_\theta(x_i)\right)^{y_i} \left(1 - h_\theta(x_i)\right)^{1-y_i}$$

- And the loglikelihood as:

$$\mathcal{L}(\theta) = \log L(\theta) = \sum_{i=1}^{n} \left[y_i \log\left(h_\theta(x_i)\right) + (1 - y_i) \log\left(1 - h_\theta(x_i)\right)\right]$$

Optimization

- Find the maximum loglikelihood of

$$\mathcal{L}(\theta) = \log L(\theta) = \sum_{i=1}^{n} \left[y_i \log\left(h_\theta(x_i)\right) + (1 - y_i) \log\left(1 - h_\theta(x_i)\right) \right]$$

- In practice, introduce the factor of '-' and convert it to find the minimum value of the

$$J(\theta) = -\log L(\theta) = -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \log\left(h_\theta(x_i)\right) + (1 - y_i) \log\left(1 - h_\theta(x_i)\right) \right]$$

Optimization

- Cost function:

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(h_\theta(x_i)\right) + (1 - y_i)\log\left(1 - h_\theta(x_i)\right)\right]$$

- $h_\theta(x_i) = \frac{1}{1+e^{-\theta^T x_i}} \Longrightarrow \log h_\theta(x_i) = -\log\left(1 + e^{-\theta^T x_i}\right)$
- $\Longrightarrow \log\left(1 - h_\theta(x_i)\right) = \log\left(1 - \frac{1}{1+e^{-\theta^T x_i}}\right) = \log\left(\frac{e^{-\theta^T x_i}}{1+e^{-\theta^T x_i}}\right) = -\theta^T x_i - \log(1 + e^{-\theta^T x_i})$
- Substitute $\log h_\theta(x_i) = -\log\left(1 + e^{-\theta^T x_i}\right)$ and $\log\left(1 - h_\theta(x_i)\right) = -\theta^T x_i - \log(1 + e^{-\theta^T x_i})$ into $J(\theta)$

$$J(\theta) - \frac{1}{n}\sum_{i=1}^{n}\left[-y_i \log\left(1 + e^{-\theta^T x_i}\right) + (1 - y_i)(-\theta^T x_i - \log(1 + e^{-\theta^T x_i}))\right]$$

Optimization

- Substitute $\log h_\theta(x_i) = -\log\left(1 + e^{-\theta^T x_i}\right)$ and

$$\log\left(1 - h_\theta(x_i)\right) = -\theta^T x_i - \log(1 + e^{-\theta^T x_i}) \text{ into } J(\theta)$$

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[-y_i \log\left(1 + e^{-\theta^T x_i}\right) + (1 - y_i)\left(-\theta^T x_i - \log\left(1 + e^{-\theta^T x_i}\right)\right)\right]$$

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[-y_i \log\left(1 + e^{-\theta^T x_i}\right) - \theta^T x_i - \log\left(1 + e^{-\theta^T x_i}\right) + y_i\theta^T x_i + y_i \log\left(1 + e^{-\theta^T x_i}\right)\right]$$

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i\theta^T x_i - \theta^T x_i - \log\left(1 + e^{-\theta^T x_i}\right)\right]$$

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i\theta^T x_i - \left(\log\left(e^{\theta^T x_i}\right) + \log\left(1 + e^{-\theta^T x_i}\right)\right)\right] \quad * \; a = \log e^a$$

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i\theta^T x_i - \log\left(1 + e^{\theta^T x_i}\right)\right] \quad ** \; \log a + \log b = \log(ab)$$

Optimization

- Cost function:

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i\theta^T x_i - \log\left(1 + e^{\theta^T x_i}\right)\right]$$

- Minimize cost function

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{\partial}{\partial \theta_j}\left(\frac{1}{n}\sum_{i=1}^{n}\left[\log\left(1 + e^{\theta^T x_i}\right) - y_i\theta^T x_i\right]\right)$$

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\partial}{\partial \theta_j}\log\left(1 + e^{\theta^T x_i}\right) - \frac{\partial}{\partial \theta_j}(y_i\theta^T x_i)\right]$$

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{1}{1 + e^{\theta^T x_i}}\frac{\partial}{\partial \theta_j}\left(e^{\theta^T x_i}\right) - y_i x_i^{(j)}\right]$$

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{x_i^{(j)} e^{\theta^T x_i}}{1 + e^{\theta^T x_i}} - y_i x_i^{(j)}\right] = \frac{1}{n}\sum_{i=1}^{n}(h_\theta(x_i) - y_i)\cdot x_i^{(j)}$$

$$\frac{\partial}{\partial \theta_j}(\theta^T x_i) =$$

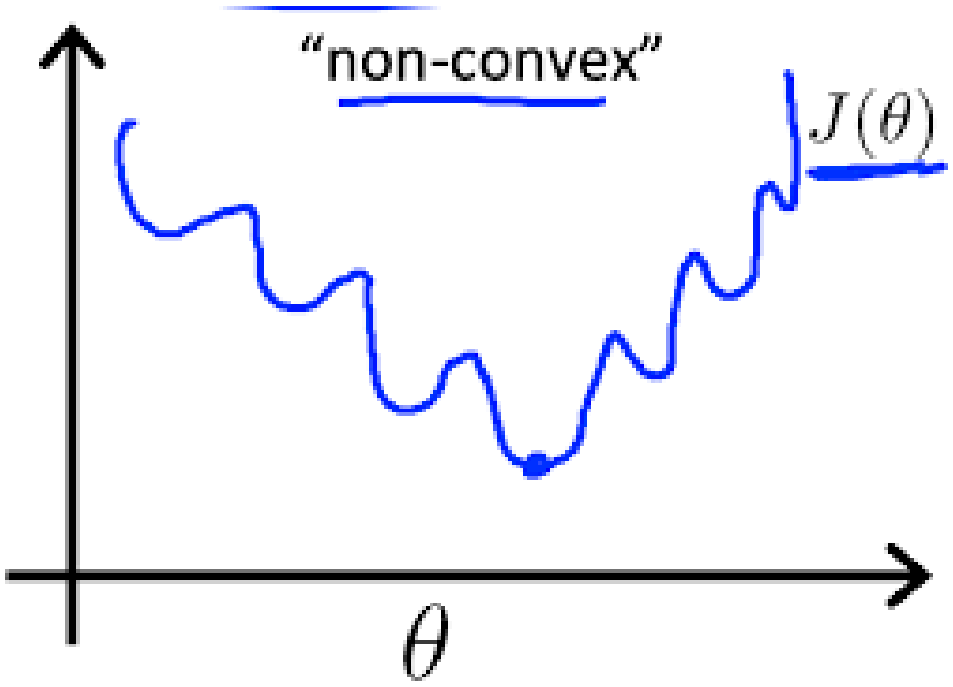
$$\frac{\partial}{\partial \theta_j}\left(\theta_1 x_i^{(1)} + \theta_2 x_i^{(2)} + \cdots + \theta_j x_i^{(j)} + \cdots\right) = x_i^{(j)}$$

$x_i^{(j)}$ is the j-th feature of x_i

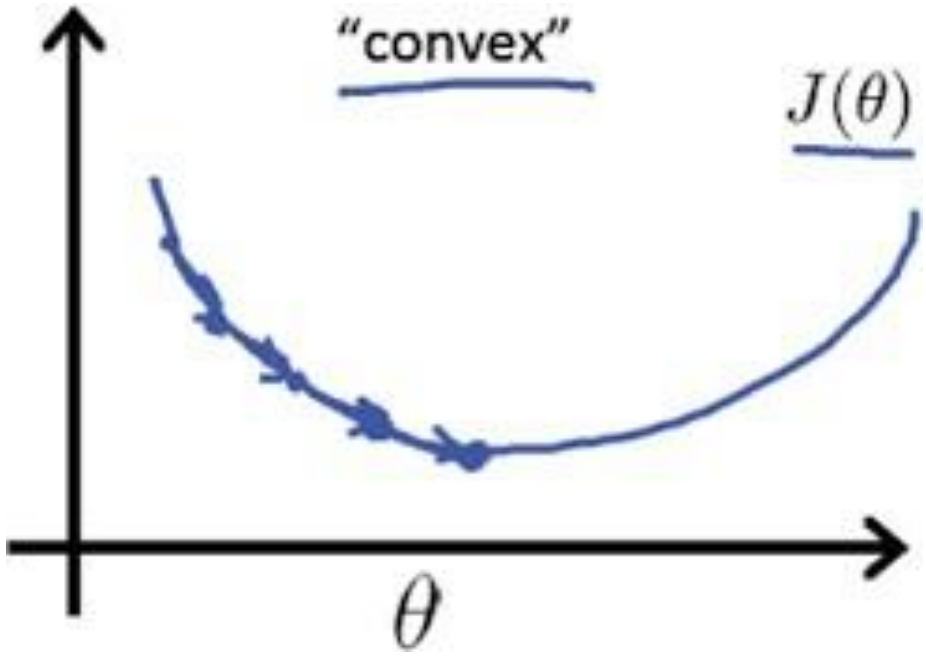
$$h_\theta(x_i) = \frac{1}{1 + e^{-\theta^T x_i}} = \frac{e^{\theta^T x_i}}{1 + e^{\theta^T x_i}}$$

Logistic regression

linear regression



logistic regression



Logistic regression

- Hypothesis:

$$g(h_\theta(x)) = \begin{cases} 1, if\, h_\theta(x) \geq 0 \\ 0, if\, h_\theta(x) < 0 \end{cases}$$

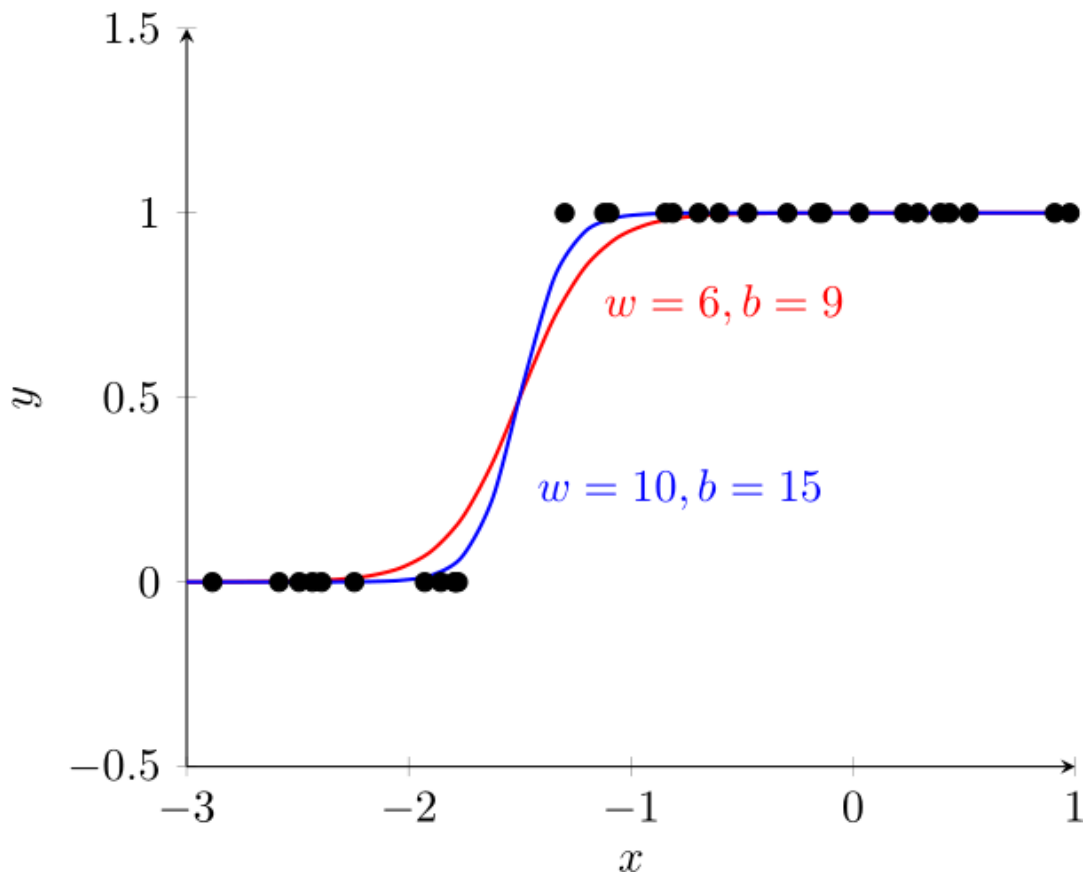
- Parameters: θ
- Cost function:

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(h_\theta(x_i)\right) + (1 - y_i)\log\left(1 - h_\theta(x_i)\right)\right]$$

- Goal:

$$\min_\theta J(\theta)$$

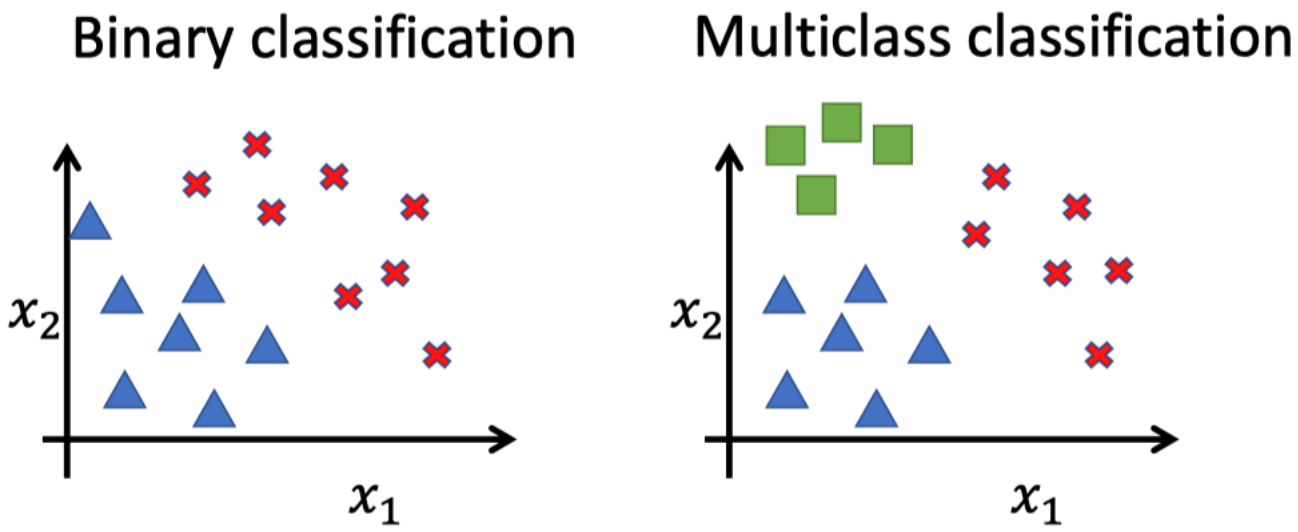
Example of uni-variable logistic regression



Multi-class Classification

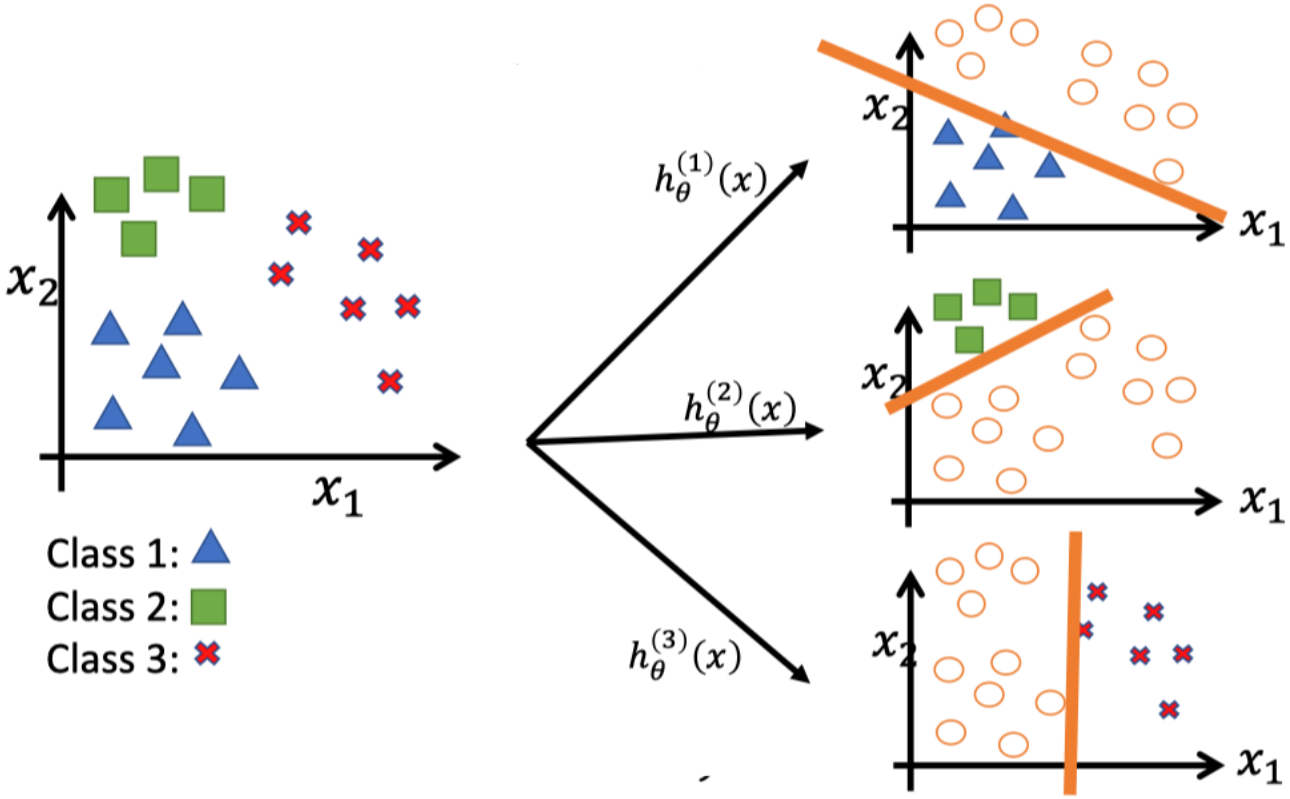
Example of logistic regression

- Logistic regression is usually only able to perform binary classifications.
- Is there a way to make logistic regression achieve multiple classifications?



One-vs-all (one-vs-rest, OvR)

- Split the training set into three separate binary classification problems
- To be classified into three categories: A, B, and C, then you can treat A as positive data, and B and C as negative data.



One-vs-all (one-vs-rest)

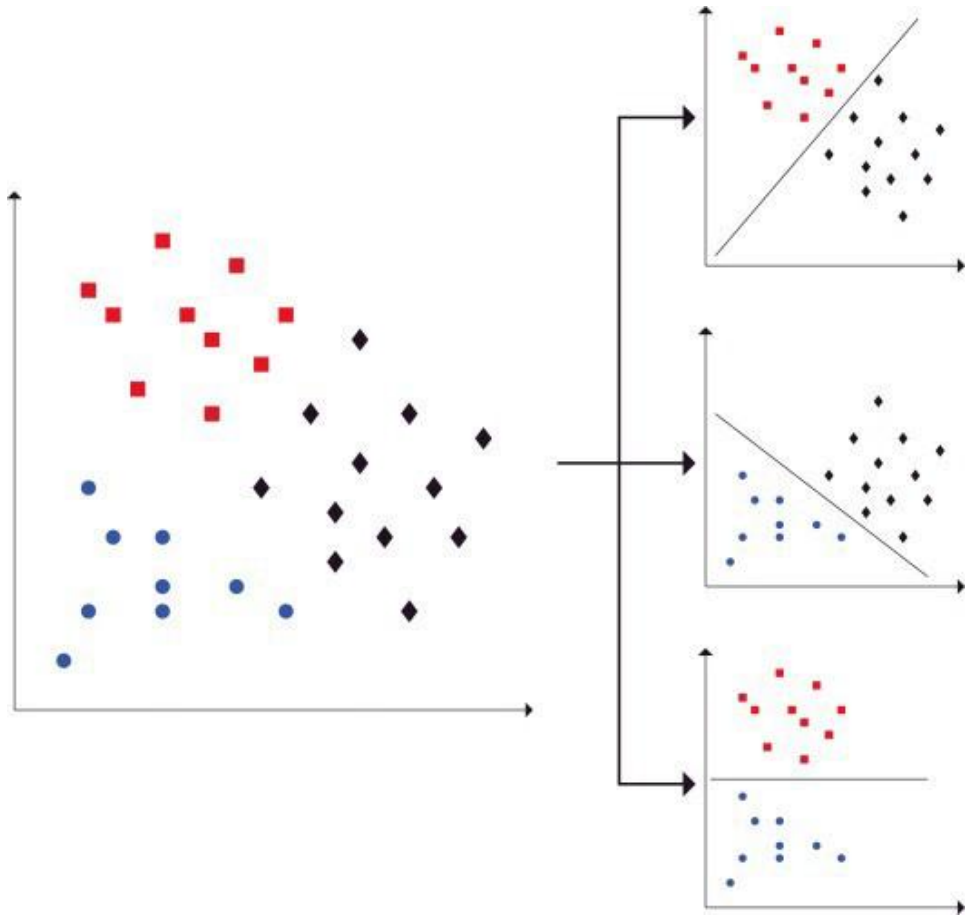
- Train a logistic regression classifier $h_\theta^{(i)}(x)$ for each class i to predict the probability that $y = i$

- Given a new input x, pick the class i that maximizes

$$\max_i h_\theta^{(i)}(x)$$

One-vs-one (OvO)

- OvO approach splits the dataset into one dataset for each class versus every other class.
- Each binary classification model may predict one class label and the model with the most predictions or votes is predicted by the one-vs-one strategy.



OvR vs. OvO

- One-vs-rest:
  - +: fast speed
  - -: low accuracy
- One-vs-one :
  - +: high accuracy
  - -: slow speed

Example of Logistic Regression

Logistic Regression Example

- Step 1: Data preprocessing
- Step 2: Construct various auxiliary functions, such as activation function, optimization function, etc.
- Step 3: Synthesize the above auxiliary functions and combine them into a model

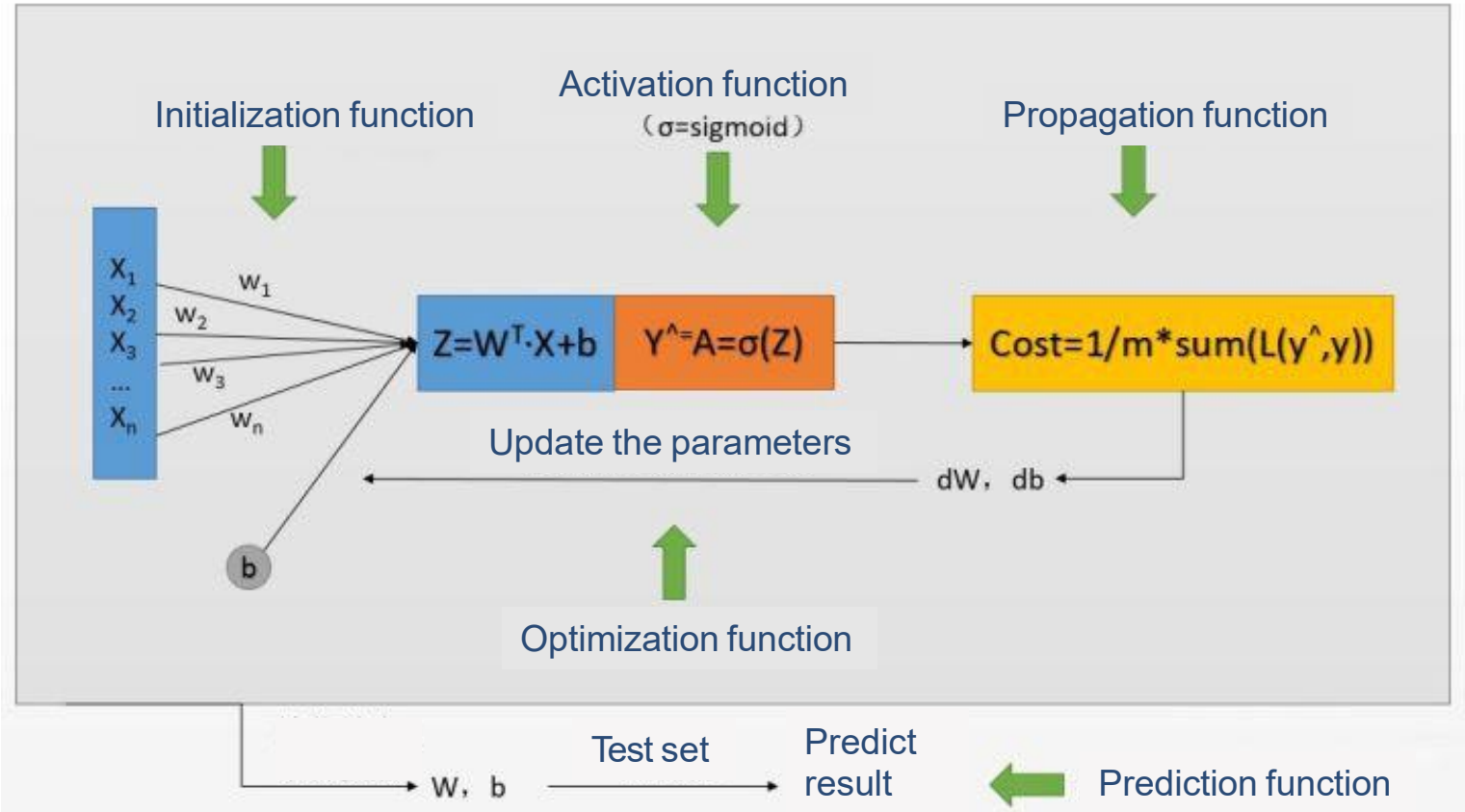
Logistic Regression Example

- Step 1: Data preprocessing
  - Figure out the shape and dimensions of the data
  - Convert data (such as pictures) into vectors (image to vector) for easy processing
  - Standardize the data for better training

Logistic Regression Example

- Step 2: Construct various auxiliary functions, such as activation function, optimization function, etc.
  - Activation function (here we use the sigmoid function)
  - Parameter initialization function (used to initialize W and b)
  - Propagation function (here is used to find loss cost and derivation of W and b, namely dW, db)
  - Optimization function (update W and b iteratively to minimize cost)
  - Prediction function (predict based on learned W and b)

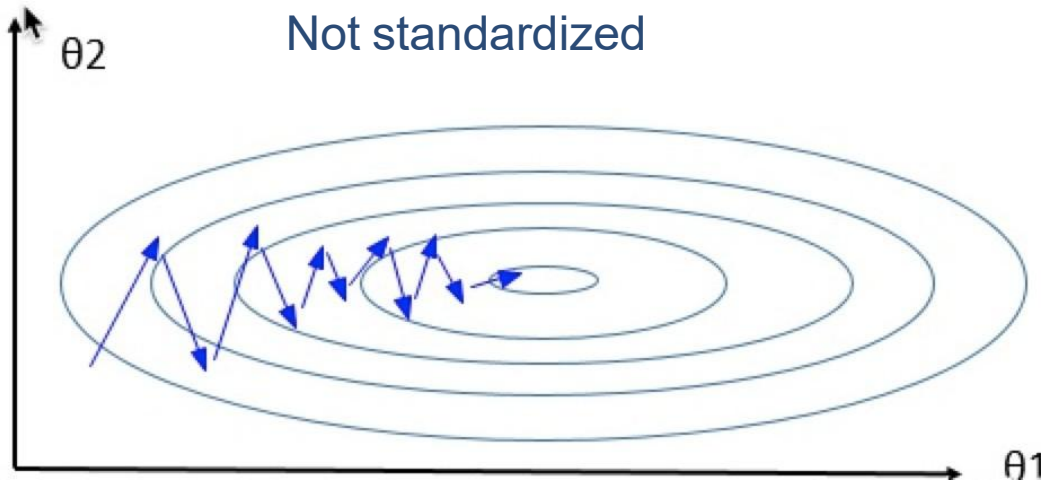
Logistic Regression Example

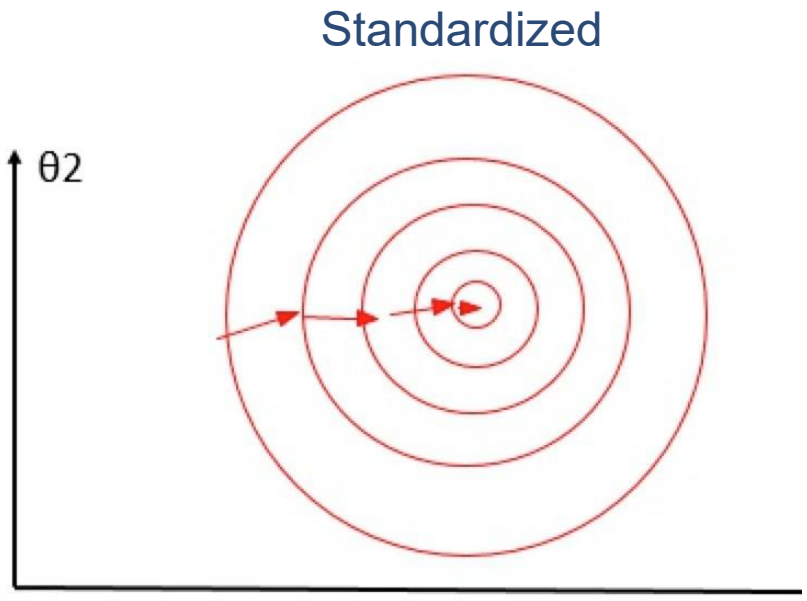


Logistic Regression Example

- Step 3: Synthesize the above auxiliary functions and combine them into a model
  - Input training set, prediction set, hyperparameters, and then give model parameters and accuracy

Standardization



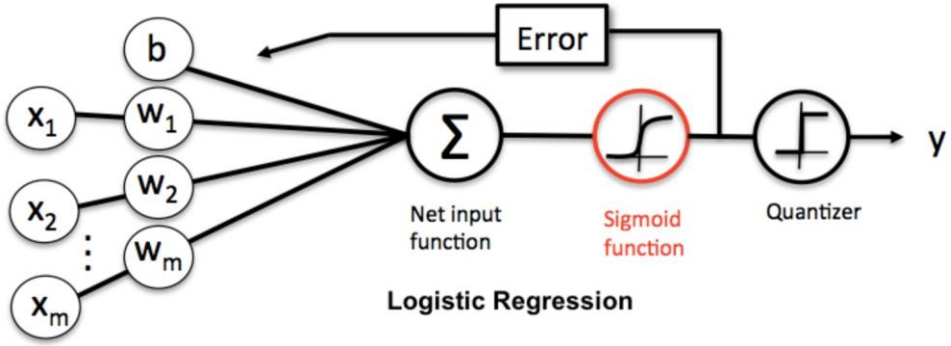


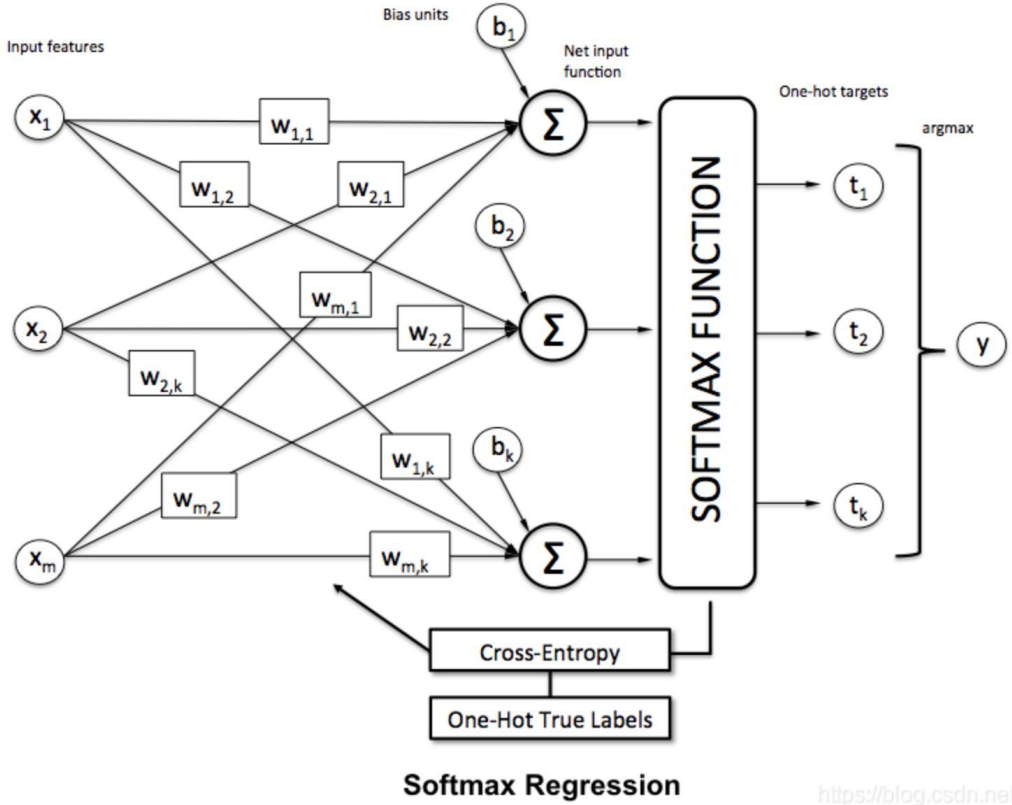
- The range of different features of the original data may vary greatly.
- The large fluctuation range causes the contour map to become very "flat", and it is easy to take detours when the gradient drops, so the gradient drops will be slower and the accuracy is not high
- After standardization (also known as normalization), the contours become regular, and the gradient is easily reduced.

Softmax Regression

Softmax Regression

- Softmax regression is a general form of logistic regression.
- When the number of categories $k = 2$, Softmax regression degenerates into logistic regression





https://blog.csdn.net/

Softmax Regression: Generalization of LR

- Logistic regression:

$$h_\theta(x) = \frac{1}{1+e^{-\theta^T x}} \implies h_\theta(x) = \frac{1}{e^{\theta_1^T x_i} + e^{\theta_2^T x_i}} \begin{bmatrix} e^{\theta_1^T x_i} \\ e^{\theta_2^T x_i} \end{bmatrix}$$

- Cost function:

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log\left(h_\theta(x_i)\right) + (1-y_i)\log\left(1-h_\theta(x_i)\right)\right]$$

- Softmax regression:

$$h_\theta(x_i) = \begin{bmatrix} p(y_i = 1|x_i;\theta) \\ p(y_i = 2|x_i;\theta) \\ \vdots \\ p(y_i = K|x_i;\theta) \end{bmatrix} = \frac{1}{\sum_{k=1}^{K} e^{\theta_k^T x_i}} \begin{bmatrix} e^{\theta_1^T x_i} \\ e^{\theta_2^T x_i} \\ \vdots \\ e^{\theta_K^T x_i} \end{bmatrix}$$

- Cost function:

$$J(\theta) = -\frac{1}{n}\left[\sum_{i=1}^{n}\sum_{k=1}^{K} \mathbb{I}\{y_i = k\} \log \frac{e^{\theta_k^T x_i}}{\sum_{l=1}^{K} e^{\theta_l^T x_i}}\right]$$

$\mathbb{I}\{y_i = k\}$: indicator function

Softmax Regression Cost Function

- Cost function

$$J(\theta) = -\frac{1}{n}\left[\sum_{i}^{n}\sum_{k}^{K} \mathbb{I}\{y_i = k\} \log \frac{e^{\theta_k^T x_i}}{\sum_{l=1}^{K} e^{\theta_l^T x_i}}\right]$$

- Derivative of cost function

$$\frac{\partial J(\theta)}{\partial \theta_k} = -\frac{1}{n}\frac{\partial}{\partial \theta_k}\left[\sum_{i}^{n}\sum_{k}^{K} \mathbb{I}\{y_i = k\} \log \frac{e^{\theta_k^T x_i}}{\sum_{l=1}^{K} e^{\theta_l^T x_i}}\right] = -\frac{1}{n}\frac{\partial}{\partial \theta_k}\left[\sum_{i}^{n}\sum_{k}^{K} \mathbb{I}\{y_i = k\}\left(\theta_k^T x_i - \log \sum_{l}^{K} e^{\theta_l^T x_i}\right)\right]$$

$$\frac{\partial J(\theta)}{\partial \theta_k} = -\frac{1}{n}\left[\sum_{i}^{n} \mathbb{I}\{y_i = k\}\left(x_i - \sum_{k}^{K} \frac{e^{\theta_k^T x_i} \cdot x_i}{\sum_{l=1}^{K} e^{\theta_l^T x_i}}\right)\right] = -\frac{1}{n}\left[\sum_{i}^{n} x_i \cdot \mathbb{I}\{y_i = k\}\left(1 - \sum_{k}^{K} \frac{e^{\theta_k^T x_i}}{\sum_{l=1}^{K} e^{\theta_l^T x_i}}\right)\right]$$

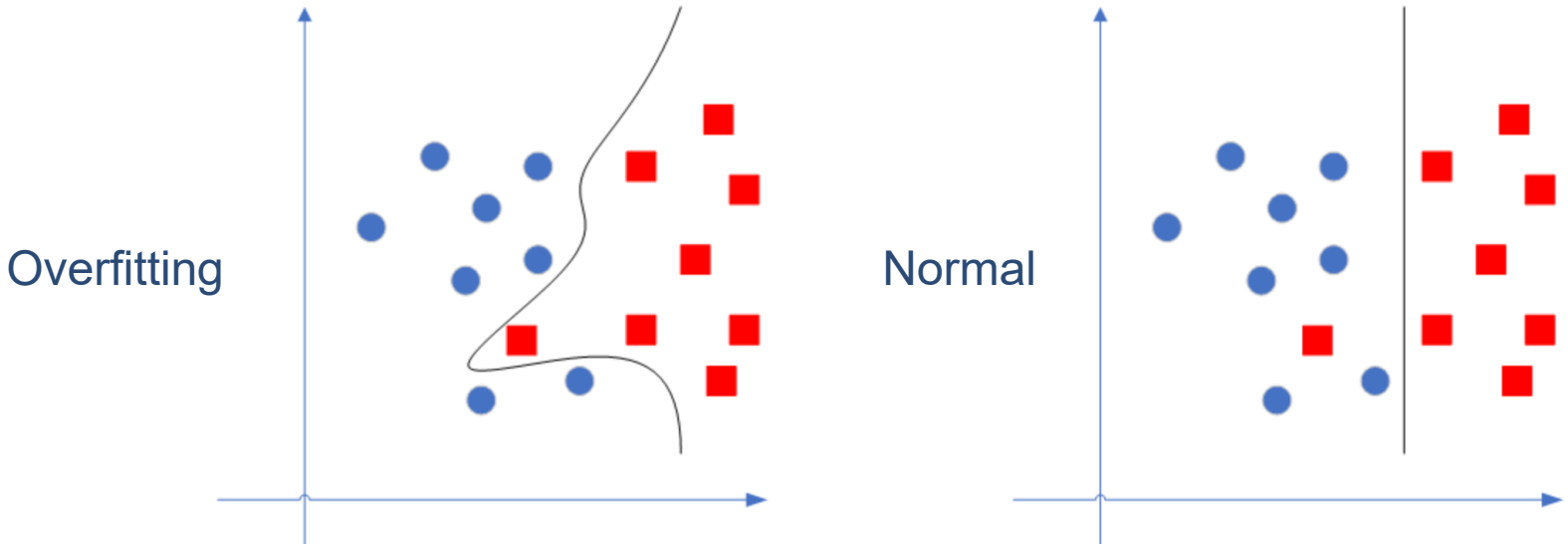
$$\frac{\partial J(\theta)}{\partial \theta_k} = -\frac{1}{n}\left[\sum_{i}^{n} x_i\left(\mathbb{I}\{y_i = k\} - \sum_{k}^{K} \mathbb{I}\{y_i = k\} \cdot \frac{e^{\theta_k^T x_i}}{\sum_{l=1}^{K} e^{\theta_l^T x_i}}\right)\right] = -\frac{1}{n}\left[\sum_{i}^{n} x_i\left(\mathbb{I}\{y_i = k\} - \frac{e^{\theta_k^T x_i}}{\sum_{l=1}^{K} e^{\theta_l^T x_i}}\right)\right]$$

$$\frac{\partial J(\theta)}{\partial \theta_k} = -\frac{1}{n}\left[\sum_{i}^{n} x_i(\mathbb{I}\{y_i = k\} - p(y_i = k|x_i;\theta))\right]$$

Regularization and overfitting

Regularization of Logistic Regression

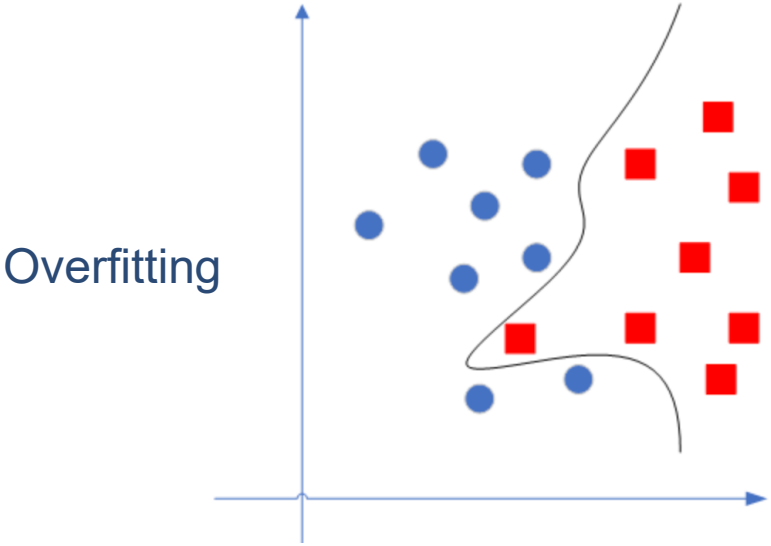
- Isn't the accuracy of the model the higher the better?

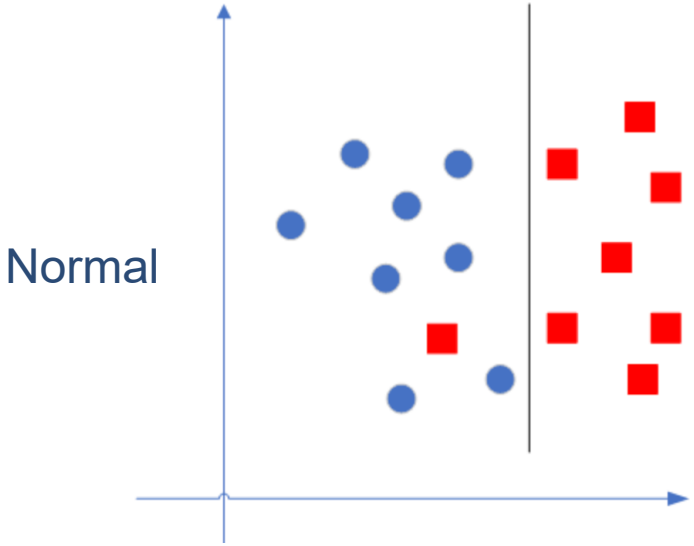


Regularization of Logistic Regression

- If the accuracy is too high, it can match the data of the training set very well, but once new data is available, it will perform very poorly

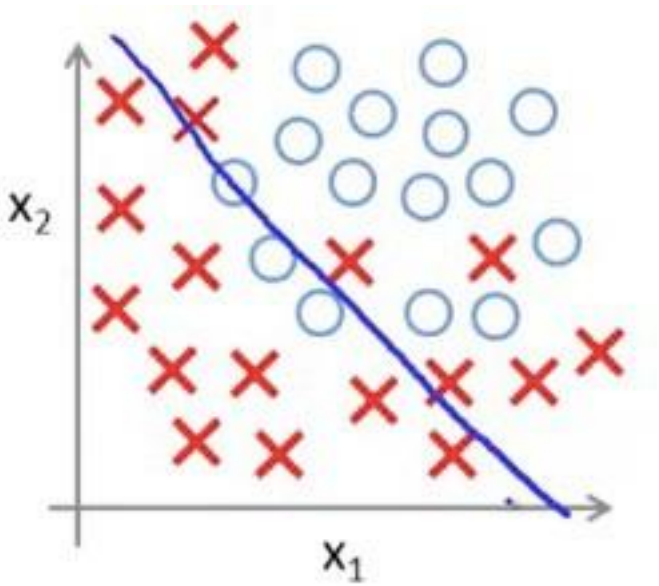
- The accuracy can be worse sometimes, but the generalization performance, which is the ability to recognize new data, is better



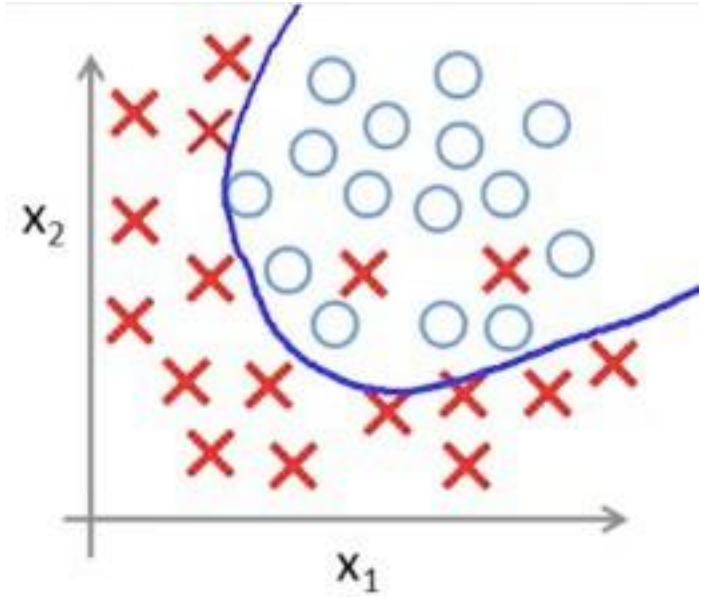


Overfitting of Logistic Regression

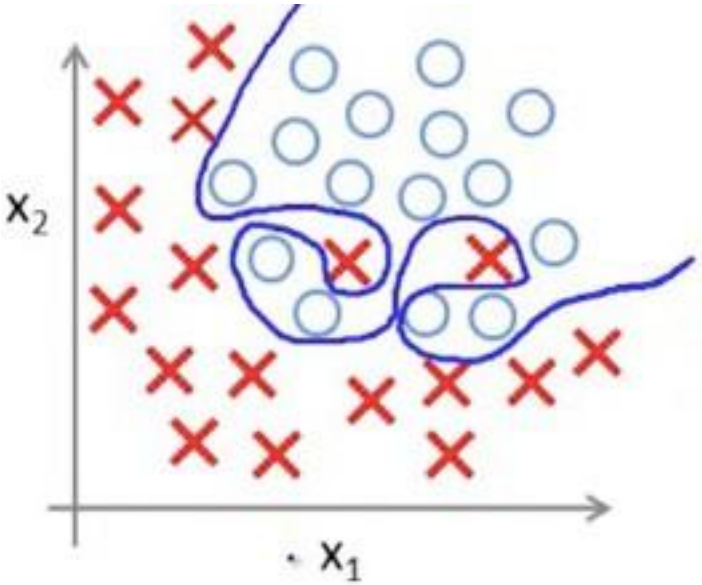
Decision boundary



$$h(\theta) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$



$$h(\theta) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta x_2^2 + \theta_5 x_1 x_2)$$



$$h(\theta) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \theta_6 x_1^3 x_2 + \ldots)$$

Softmax Classifier

Want to interpret raw classifier scores as **probabilities**

$$s = f(x_i; W)$$

$$P(Y = k | X = x_i) = \frac{e^{s_k}}{\sum_j e^{s_j}}$$

Softmax Function

cat	**3.2**
car	5.1
frog	-1.7

Softmax Classifier (Multinomial Logistic Regression)

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Probabilities must be >= 0

cat	**3.2**	exp →	**24.5**
car	5.1		164.0
frog	-1.7		0.18

unnormalized probabilities

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Softmax Function

Probabilities must be >= 0

Probabilities must sum to 1

cat	**3.2**	exp →	**24.5**	normalize →	0.13
car	5.1		164.0		0.87
frog	-1.7		0.18		0.00

unnormalized probabilities

probabilities

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exp → normalize →

Unnormalized log-probabilities / logits

unnormalized probabilities

probabilities

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Probabilities must sum to 1

$$L_i = -\log P(Y = y_i | X = x_i)$$

	Unnormalized log-probabilities / logits	exp →	unnormalized probabilities	normalize →	probabilities
cat	**3.2**		**24.5**		**0.13**
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→ L_i = -log(0.13)
= 2.04

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car	5.1		164.0		0.87
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→ L_i = -log(0.13)
= 2.04

Maximum Likelihood Estimation
Choose weights to maximize the likelihood of the observed data

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	Unnormalized log-probabilities / logits	exp →	unnormalized probabilities	normalize →	probabilities	compare	Correct probs
cat	**3.2**		**24.5**		**0.13**		**1.00**
car	5.1		164.0		0.87		0.00
frog	-1.7		0.18		0.00		0.00

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	Unnormalized log-probabilities / logits	exp →	unnormalized probabilities	normalize →	probabilities	compare	Correct probs
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Cross Entropy

$$H(P, Q) = H(p) + D_{KL}(P \| Q)$$

Softmax Classifier (Multinomial Logistic Regression)

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Want to interpret raw classifier scores as **probabilities**

$$\boxed{s = f(x_i; W)}$$

$$\boxed{P(Y = k | X = x_i) = \frac{e^{s_k}}{\sum_j e^{s_j}}}$$

Softmax Function

Maximize probability of correct class

$$L_i = -\log P(Y = y_i | X = x_i)$$

Putting it all together:

$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

Softmax Classifier (Multinomial Logistic Regression)

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Putting it all together:

$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

Q1: What is the min/max possible softmax loss L_i?

A: min 0, max infinity

Q2: At initialization all s_j will be approximately equal; what is the softmax loss L_i, assuming C classes?

Softmax Classifier (Multinomial Logistic Regression)

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Maximize probability of correct class

$$L_i = -\log P(Y = y_i | X = x_i)$$

Putting it all together:

$$L_i = -\log\left(\frac{e^{s_{y_i}}}{\sum_j e^{s_j}}\right)$$

Q2: At initialization all s will be approximately equal; what is the loss?
A: -log(1/C) = log(C),
If C = 10, then L_i = log(10) ≈ 2.3

Implementation

Softmax Regression

- from Scratch
- https://d2l.ai/chapter_linear-networks/softmax-regression-scratch.html
- **Concise implementation**
- https://d2l.ai/chapter_linear-networks/softmax-regression-concise.html